5.3.1 Simple harmonic oscillations

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	displacement, amplitude, period, frequency, angular frequency and phase difference	*M4.7*
(b)	angular frequency ω; $\omega = \frac{2\pi}{T}$ or $\omega = 2\pi f$	
(c)	**(i)** simple harmonic motion; defining equation $a = -\omega^2 x$	
	(ii) techniques and procedures used to determine the period/frequency of simple harmonic oscillations	**PAG10** e.g. mass on a spring, pendulum
(d)	solutions to the equation $a = -\omega^2 x$	*M3.9, M3.12*
	e.g. $x = A\cos\omega t$ or $x = A\sin\omega t$	
(e)	velocity $v = \pm\omega\sqrt{A^2 - x^2}$ hence $v_{max} = \omega A$	*M2.2*
(f)	the period of a simple harmonic oscillator is independent of its amplitude (isochronous oscillator)	
(g)	graphical methods to relate the changes in displacement, velocity and acceleration during simple harmonic motion.	